Training-Free Framework for Single-Turn Web Intent Understanding with Retrieval-Augmented Reasoning and Calibrated Refusal

INPUT: a single-turn user query q

(context-sparse, web search / IM)

q (example)

"How do I change my email password on Gmail?"

External Intent Knowledge Base (KB, zero-annotation, updatable at inference)

Intent Prototypes (centroid embeddings)

Change Password · Account Settings · Payment Issue · ... · Other Intents

Utterance Vectors (examples per intent)

"How to reset my Gmail password?"	Change Password
"I forgot my email login, help!"	Change Password
"Update my recovery email."	Account Settings
...	

Module 1: Retrieval & Reasoning

1) Top-K intent retrieval from KB

Change Password	Account Settings	Payment Issue	...
0.92	**0.64**	**0.41**	Top-K (e.g., K=10)

2) LLM self-assessed reasoning & confidence

Candidate intents (by rank)	Conf_LLM
1. Change Password (primary)	**0.86**
2. Account Settings (secondary)	**0.42**
3. Payment Issue	**0.17**
...	

Outputs: candidate intents with self-assessed confidence Conf_LLM (primary, secondary)

Module 2: Set-Valued Conformal Calibration

1) Rank-normalized score fusion

$$s(q) = g(\text{stats_ret}, \text{Conf_LLM})$$

Top-K similarity · Margin (top1-top2) · Distance (min dist.) · Conf (LLM/top1)

2) Split conformal calibration (no gradients)

Calibration Split (held-out) → Nonconformity score s(q) → τ_cal → Finite-sample coverage 1 − α = 0.90

3) Theoretical guarantees

Marginal coverage: $\Pr(Y \in \hat{S}(q)) \geq 1-\alpha = 0.90$

Set-level Jaccard risk ≤ ε_J (finite-sample bound)

No gradient training, no classifier head

Module 3: Adaptive Three-State Decision

Thresholds calibrated at FPR95

Low Risk ... High Risk

−∞ · τ_lo · s(q) · τ_hi · +∞

RELEASE

s(q) < τ_lo → **Output: 1-intent set** — Downstream routing

CLARIFY

τ_lo ≤ s(q) ≤ τ_hi → **Output: 2–3 intents** — User confirmation

REJECT

s(q) > τ_hi → **Output: empty set** — Human escalation

OPTIONAL COMPARISON (Not Required)

Light Fine-Tuning (LoRA / SetFit) — **NOT Required** → **Retrieval + Calibration + Decision Only** with Theoretical Guarantees

Finite-sample coverage guarantee: $P(Y \in \hat{S}(q)) \geq 1-\alpha$ under exchangeability · No gradient updates · Black-box LLM compatible